

The low oxalate diet is a dietary approach that focuses on reducing the intake of foods high in oxalates, which are naturally occurring compounds found in many plants and some animal-based foods. Oxalates, or oxalic acid, are naturally occurring compounds that plants use as a defense mechanism. They are present in various foods, including vegetables, fruits, nuts, seeds, and grains. Oxalates are not inherently harmful to most people, but for those who are prone to kidney stones, they can contribute to the formation of calcium oxalate stones. By reducing oxalate intake, the low oxalate diet helps to lower the risk of stone formation and can improve overall kidney health. Some patients report benefit from a low oxalate diet with a Hx of Fibromyalgia characterized by widespread pain, fatigue, sleep disturbances, cognitive difficulties and gastrointestinal issues. Oxalates have been hypothesized to contribute to fibromyalgia symptoms in some cases, particularly due to their potential impact on inflammation, pain, and muscle function.

1. Oxalates and Muscle Pain

One of the key symptoms of fibromyalgia is muscle pain, and there is some evidence suggesting that oxalates may contribute to this pain in certain individuals. When oxalates bind to calcium, they can form calcium oxalate crystals. In people with a sensitivity or an over-accumulation of oxalates, these crystals can potentially deposit in tissues, including muscles, joints, and connective tissues, leading to irritation, inflammation, and pain. Some people with fibromyalgia have reported improvements in muscle pain and fatigue after reducing their intake of oxalate-rich foods.

2. Oxalates and Mitochondrial Dysfunction

There is growing interest in how mitochondrial dysfunction may play a role in fibromyalgia. Mitochondria are responsible for producing energy in cells, and some research has suggested that mitochondrial dysfunction may contribute to the fatigue and muscle pain associated with fibromyalgia. Oxalates can interfere with mitochondrial function by binding to essential minerals like magnesium and calcium, which are needed for proper cellular energy production. When oxalates bind to these minerals, they reduce their availability for biological processes, potentially leading to issues with energy metabolism and contributing to the fatigue and muscle dysfunction commonly seen in fibromyalgia.

3. Oxalates and Inflammation

Fibromyalgia is often associated with heightened sensitivity to pain and chronic low-level inflammation. Oxalates may contribute to inflammation by promoting the formation of calcium oxalate crystals in tissues, which can trigger an inflammatory response. This inflammation could exacerbate pain and discomfort in people with fibromyalgia, particularly in soft tissues, joints, and muscles.

4. Oxalates and Gastrointestinal Health

Fibromyalgia is frequently associated with gastrointestinal symptoms, including irritable bowel syndrome (IBS), bloating, and other digestive issues. Some research suggests that oxalates can irritate the gastrointestinal tract, especially in individuals who have compromised gut health, such as those with leaky gut syndrome, IBS, or other digestive disorders.

Oxalates in Food

Here is a comprehensive list of high oxalate foods to consider avoiding or limiting if you're aiming to reduce your oxalate intake. Foods with high oxalate content generally contain more than 10 mg of oxalate per serving.

High Oxalate Foods

Vegetables

- **Spinach** (one of the highest in oxalates)
- **Beets** and beet greens
- **Swiss chard**
- **Rhubarb**
- **Okra**
- **Sweet potatoes**
- **Eggplant**
- **Collard greens**
- **Turnip greens**
- **Bok choy**
- **Kale (Curly)**
- **Zucchini (with skin)**
- **Parsnips**
- **Leeks**
- **Bell peppers**
- **Endive**

Fruits

- **Raspberries**
- **Blackberries**
- **Strawberries**
- **Kiwi**
- **Figs** (dried and fresh)
- **Oranges**
- **Grapefruit**
- **Tangerines**
- **Clementines**
- **Currants**
- **Lemons and limes** (high in peel, low in juice)
- **Plums**
- **Pineapples**
- **Dates** (especially dried)
- **Mangoes**
- **Avocados** (moderate oxalates)

Nuts and Seeds

- **Almonds**
- **Cashews**
- **Pecans**
- **Walnuts**
- **Peanuts**
- **Pine nuts**

- **Hazelnuts**
- **Macadamia nuts**
- **Sesame seeds**
- **Sunflower seeds**
- **Pistachios**

Grains and Grain Products

- **Wheat bran**
- **Whole wheat flour**
- **Buckwheat**
- **Quinoa**
- **Amaranth**
- **Barley**
- **Brown rice**
- **Corn grits**
- **Oat bran**

Legumes and Beans

- **Soy products** (tofu, soy milk, tempeh, soy sauce)
- **Navy beans**
- **Black beans**
- **Kidney beans**
- **Pinto beans**
- **Lentils**
- **Chickpeas** (Garbanzo beans)
- **Lima beans**
- **Split peas**

Beverages

- **Black tea** (especially strong or iced tea)
- **Green tea**
- **Herbal teas made with high-oxalate herbs** (e.g., starfruit, hibiscus)
- **Cocoa and hot chocolate**
- **Dark beer**
- **Soy milk**
- **Cola** (moderate oxalates)

Other Foods

- **Chocolate** (dark and milk)
- **Cocoa powder**
- **Carob**
- **Tahini**
- **Potatoes (with skin)**
- **French fries** (from potatoes with skin)
- **Miso** (soy-based)
- **Cinnamon**
- **Ginger**

Condiments and Spices

- **Black pepper**
- **Cumin**
- **Cinnamon**
- **Turmeric**
- **Coriander**
- **Oregano**

Dairy Alternatives

- **Almond milk**
- **Soy milk**
- **Rice milk**

Moderately High Oxalate Foods (to Consume Sparingly)

These foods contain oxalates but are not as high as the foods listed above.

- **Carrots**
- **Celery**
- **Tomatoes**
- **Apples**
- **Blueberries**
- **Potatoes (without skin)**
- **Peaches**
- **Grapes**
- **Broccoli**
- **Cabbage**

Low-Oxalate Alternatives

To replace high-oxalate foods, consider:

- **Low-oxalate vegetables:** Cauliflower, iceberg lettuce, cucumbers, zucchini (without skin), and cabbage
- **Low-oxalate fruits:** Bananas, apples (peeled), melons, grapes, cherries, and pineapples
- **Nuts and seeds:** Macadamia nuts, flaxseeds (in moderation), and pumpkin seeds

By managing your intake of these high oxalate foods, you can effectively reduce oxalate levels in your diet, which may help manage health issues such as kidney stones. Always consult with a healthcare professional before making significant dietary changes.